

ROVIO: Simulation, Control and Vision-Based Autonomy for a Student-Built Underwater Vehicle

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Abstract—ROVIO is an eight-thruster underwater vehicle built by a student club. This paper reports my contributions to it, from the sealing of its enclosure to a simulated autonomy stack. A flange seal with an EPDM gasket replaced radial O-ring seals, which could not be made tight on an out-of-round tube. The vehicle was modelled in the Stonefish simulator and placed in a generated coral reef. A cascaded controller with thruster allocation holds depth to 0.4 cm and tracks commanded rates within 10 %; identification showed the effective inertia of the simulated plant to be about seven times the first estimate, and the roll loop to be bounded by a loop delay of about 90 ms. Localization from fiducial markers reaches a mean position error of 0.05 m at 4 m, and I show that the lateral uncertainty of a single marker grows with the cube of the range and is not reduced by an attitude prior. A monocular pipeline, trained on automatically labelled simulator images, mapped 50 of 53 corals in view with a mean error of 0.20 m on unattended survey flights, and eleven of thirteen corals it had not seen. Testing also exposed an uninitialised-memory fault in the simulator that corrupted the vehicle state in 17 of 20 starts. All software results were obtained in simulation with ground-truth vehicle state.

Index Terms—underwater robotics, thruster allocation, cascaded control, fiducial localization, monocular mapping, synthetic training data, Stonefish

I. INTRODUCTION

ROVIO is the underwater vehicle of the Robotics and Machine Intelligence club of the National Institute of Technology Tiruchirappalli. Its first version leaked and its frame flexed; the second was a deliberately simple four-thruster vehicle; the third, the subject of this paper, has eight thrusters and control authority in six degrees of freedom (Fig. 1). A student team cannot develop control and perception software on a vehicle that is still being machined, nor afford to discover sign errors and unstable loops in a pool. The work reported here therefore had two aims: to keep water out of the enclosure with the means of a student workshop, and to provide a simulated vehicle on which the software can be developed, measured and tested before the hardware is ready.

The contributions are:

- an experimental comparison of radial and flange seals for a low-pressure enclosure made to workshop tolerances (Section III);
- a Stonefish model of the vehicle and an open-source generator of coral reef scenes (Section IV);
- a cascaded controller with thruster allocation, with an identification of the simulated plant that revises the design assumptions (Section V);

- an observability analysis of an inertial and pressure-based state estimator (Section VI);
- marker-based localization with a verified covariance, and a closed-form cube law for its weak direction (Section VII);
- a monocular coral-mapping pipeline with automatic labelling, evaluated on held-out corals, and a survey guidance layer (Sections VIII and IX);
- the diagnosis and correction of a start-up fault in the simulator (Section X).

The vehicle is the work of a team; the mechanical design of the third version and its sealing as built are by the mechanical group. The sealing study and all software described here are my own work.

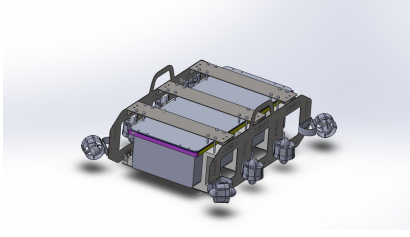
II. RELATED WORK

Open simulators for marine robotics include the Gazebo-based UUV Simulator [1] and DAVE [2], which are parametrised by hydrodynamic coefficients, and HoloOcean [3]. Stonefish [4] derives buoyancy, drag and added mass from the vehicle geometry and renders the attenuation of light in water [5], which suits a team without tow-tank data and a perception task that depends on colour. Marine vehicle models and model-based controllers are standard [6], [7], but they require identified coefficients; open autopilots such as PX4 [8] and ArduSub instead use a cascade of a proportional pose loop around a PID rate loop [9], which is adopted here as a baseline. The distribution of a wrench over redundant thrusters is the control allocation problem [10]. Fiducial markers such as AprilTag [11] and ArUco [12] have been evaluated under water [13], and the two-fold ambiguity of planar pose estimation is well known [14], [15]. Tracking by detection [16], [17] on a YOLO-family detector [18], [19], bearing-only triangulation [20], [21] and appearance randomisation of simulated training data [22] are the building blocks of the perception pipeline.

III. ENCLOSURE SEALING

The enclosure needs a static seal that can be opened routinely and is tight at the depth of the test pool, about 3 m, where the hydrostatic pressure is only 29 kPa. The difficulty is therefore not strength but a continuous line of contact between parts that a student workshop can make.

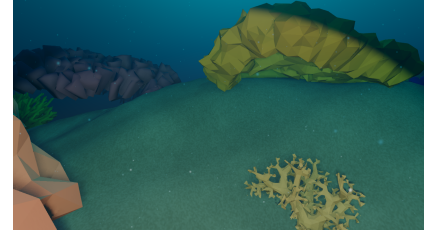
A radial seal, with an O-ring in a groove around a plug, was tested first on printed plugs of graded diameters. It did



(a) CAD model of the vehicle

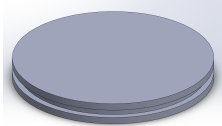


(b) Vehicle above the generated reef

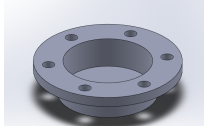


(c) Forward camera at 12 m depth

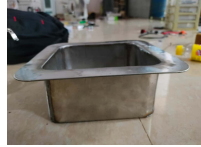
Fig. 1. The third version of ROVIO and its simulation in Stonefish. The reef is produced by a generator from a layout file; coral colours are chosen so that a hue survives the absorption of red light at depth.



(a) O-ring test disc



(b) Flange test rig



(c) Steel test box

Fig. 2. Sealing study: a printed test piece for the radial seal, the rig on which the flange seal was tested, and the box used to verify it before use on the vehicle.

not become tight. An O-ring seals only where its squeeze is positive all the way round, and the gap it has to close varies around a bore that is not round: a plug large enough to squeeze the ring at the widest part of the acrylic tube no longer entered at the narrowest. The low pressure does not help, since an O-ring relies on pressure to seat it. A flange seal was then tested on a dedicated rig and sealed reliably, for the converse reason: a flat gasket is compressed axially by a ring of bolts, so the contact pressure is set by the bolts and not by the fit of two diameters, and the broad annular area tolerates local defects. EPDM was selected as the gasket material because it is softer than the nitrile of standard O-rings and conforms to a printed or machined face at a lower bolt load. The tests also showed that printed PLA walls pass water between layers and must be surface-finished. The lid of the current enclosure uses this seal; it was verified on a steel test box of the same form held under water (Fig. 2).

IV. SYSTEM AND SIMULATION

The software is a set of ROS 2 [23] packages that communicate through topics (Fig. 3). The interface between autonomy and control is a body-frame velocity command with a timeout of 0.5 s, on which the controller holds its pose; every failure of a higher layer therefore ends in a hold. The marine conventions are used throughout [6]: a north-east-down world frame and a forward-starboard-down body frame.

A. Vehicle model

The CAD model was exported as a URDF description and converted to the Stonefish scenario format. The exporter assumes an east-north-up frame and Stonefish a north-east-down one, so every pose, centre of mass and inertia tensor was

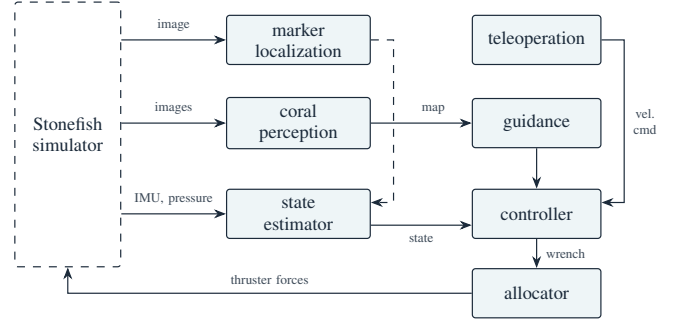


Fig. 3. Software architecture. Each box is a ROS 2 package. In the reported results the state is the simulator's ground truth; the estimator and its marker input (dashed) are the path that replaces it.

re-expressed ($\mathbf{I}' = \mathbf{T}\mathbf{I}\mathbf{T}^T$), and the meshes were decimated from more than 600 000 triangles. The model has a mass of 14.73 kg and eight ideal thrusters of ± 100 N; it carries an inertial sensor, a pressure sensor, a forward camera pitched 20° down and a downward camera (1280×720 pixels, 80° field of view).

B. Reef generator

A generator, published as a repository of its own, assembles a reef from a layout file and 33 procedurally generated coral models of nine families. The layout file is the only place in which a coral position is written: the scene is generated from it and the perception software reads the positions back as ground truth. Marker positions and camera mountings are likewise defined once and shared by the scene and the software, since a disagreement of a few degrees would bias every vision result without a visible symptom. Collision meshes of about 3 000 triangles are also used for display, which reduces the scene of 111 corals from 4.2 to 0.47 million triangles.

C. Light at depth

In the first runs every coral appeared grey-green. Stonefish attenuates light along its path and red is absorbed first, as in the sea. The lamps were made brighter, and the generator gained an option that writes saturated colours with an amplified red component, so that a coral 3 m away shows its hue (Fig. 1c). Colour is still lost at 8 m, deliberately: a detector has to cope with the same effect in the sea.

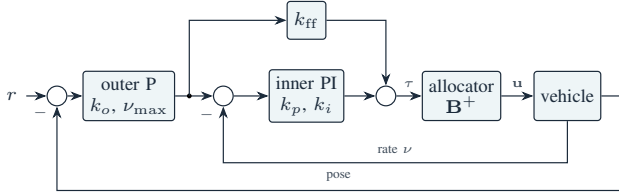


Fig. 4. Control cascade of one degree of freedom. The outer loop turns the pose error into a limited rate setpoint; the inner loop, with drag feed-forward, turns the rate error into a force or torque.

V. CONTROL

A. Thruster allocation

With thruster i at $\mathbf{r}_i$ thrusting along $\mathbf{e}_i$, the wrench produced by the thruster forces $\mathbf{u} \in \mathbb{R}^8$ is

$$\boldsymbol{\tau} = \mathbf{B}\mathbf{u}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_8 \\ \mathbf{r}_1 \times \mathbf{e}_1 & \cdots & \mathbf{r}_8 \times \mathbf{e}_8 \end{bmatrix}. \quad (1)$$

Four thrusters at 45° in the horizontal plane produce surge, sway and yaw, and four vertical ones heave, roll and pitch. $\mathbf{B}$ has rank 6 and a condition number of 9.76. Its singular values, 2.00, 1.41, 1.41, 0.85, 0.49 and 0.20, each belong to one axis, and pitch is the weakest because the vertical thrusters stand only 0.10m ahead of and behind the centre. The allocator uses the pseudo-inverse $\mathbf{u} = \mathbf{B}^\top(\mathbf{B}\mathbf{B}^\top)^{-1}\boldsymbol{\tau}$, the solution of least squared thrust, and on saturation scales all eight forces by a common factor. This preserves the direction of the wrench at the price of reducing attitude torque together with thrust; a constrained allocation [10] is the remedy.

B. Cascaded controller

Each degree of freedom has two loops (Fig. 4). For a pose error e and a measured rate ν ,

$$\nu_{\text{sp}} = \text{sat}_{\nu_{\text{max}}}(k_o e), \quad (2)$$

$$\boldsymbol{\tau} = \text{sat}_{\tau_{\text{max}}}\left(k_p(\nu_{\text{sp}} - \nu) + k_i \int (\nu_{\text{sp}} - \nu) dt + k_{\text{ff}} \nu_{\text{sp}}\right), \quad (3)$$

with a clamped integrator. The feed-forward term supplies the drag expected at the commanded rate, and a constant force F_b carries the net weight. A non-zero command bypasses the outer loop and moves the pose setpoint to the present pose, so the vehicle moves while a stick is deflected and holds when it is released. Four modes (manual, stabilise, depth hold, position hold) close progressively more loops.

C. Design, identification and retuning

Each axis was treated as a first-order system $m\dot{\nu} + d\nu = \tau$. The outer gain follows from the desired settling time, $T_o = 1/k_o$, and the inner gain from the requirement that the inner loop, with time constant $T_i = m/(k_p + d)$, be about three times faster. First gains were calculated with the rigid-body mass increased for added mass, 24 kg, and an assumed net weight of 5 N. They were inadequate: a dive to 1 m overshoot by 0.42 m, and the rate loops settled at a third to a half of their setpoints.

TABLE I
IDENTIFIED PARAMETERS OF THE SIMULATED PLANT

Quantity	Assumed	Measured
Net weight	5 N	75 N
Effective mass, surge	24 kg	180 kg
Yaw inertia	0.4 kg m ²	17 kg m ²
Drag, surge and sway	n/a	120 N/(m/s)
Drag, heave	n/a	280 to 350 N/(m/s)
Drag, yaw	n/a	15 N m/(rad/s)

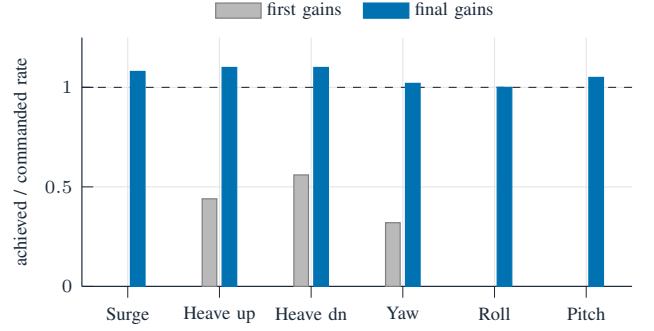


Fig. 5. Rate reached at full stick as a fraction of the commanded rate, before and after identification and retuning (single runs; no comparable first-gain figure exists for surge, roll and pitch).

The plant was therefore identified by step tests with a purpose-written tool (Table I). The vehicle sinks with 75 N, not 5 N, so the depth integrator had been spending most of its authority on holding the vehicle up. The effective mass is about 180 kg, twelve times the dry mass, because the simulator derives added mass from a shape fitted to an open frame. And the drag on the linear axes and in yaw is large compared with the first gains, so that a proportional loop settles at the fraction $k_p/(k_p + d)$ of its setpoint. The gains were revised accordingly: the net weight and the measured drag became feed-forward terms, and the inner gains were raised by step tests by factors of 1.5 to 10. Every axis now reaches its commanded rate within 10 % (Fig. 5), and in depth hold the vehicle stays within 0.4 cm, 0.03° of roll, 0.83° of pitch and 0.31° of heading over 30 s.

D. The roll loop is bounded by latency

Roll has the smallest inertia and almost no damping. With the inner gains raised from 2 to 3 the roll rate showed a ripple of $5.9^\circ/\text{s}$ in hover, and at 4 the loop entered a limit cycle at 2.8 Hz. The plant of the rate loop is close to an integrator, which contributes 90° of phase lag, so the loop oscillates where its delays contribute the other 90° : a delay of about 90 ms, between four and five periods of the 50 Hz loop. The roll gain is therefore bounded by latency and not by thrust, and it must be established again on the hardware. In position hold the weak roll loop is the reason the roll target of 2° is exceeded (2.6°), since sway couples into roll.

VI. STATE ESTIMATION

The vehicle will carry an inertial sensor, a pressure sensor and cameras, and no Doppler velocity log. An extended Kalman

filter [24] with a generic constant-acceleration model fuses attitude, angular rate, gravity-compensated acceleration and depth. Three groups of states must be distinguished. Depth, roll and pitch are measured directly and without drift. Heading drifts with the inertial sensor, at about 3° per minute in the scene. Horizontal position and velocity are unobservable from these sensors: an attitude error δ leaves a component $g\delta$ of gravity in the horizontal acceleration, and for $\delta = 0.01$ rad this integrates to 1.2 m in 5 s.

With the loops closed on the estimate, and a noisy position stand-in of 0.10 m at 5 Hz, depth was held to 0.8 cm and roll and pitch within their targets in depth hold, so for these states the estimate can replace ground truth today. Heading missed its target through drift alone, and position hold wandered by 19 cm with 3.5° of roll, because the velocity that the filter infers from noisy fixes drives a controller that was tuned on a noise-free one. The missing pieces are an absolute heading and position reference, which Section VII provides, and a velocity source or a model-based prediction to bridge the gaps between fixes.

VII. MARKER LOCALIZATION

Fourteen boards with AprilTag markers of side $s = 0.6$ m stand at surveyed positions around the reef. With $\mathbf{X}_j$ a surveyed corner, $\mathbf{x}_j$ its detected pixel and π the pinhole projection for a vehicle pose ξ , the pose minimises

$$E(\xi) = \sum_j \frac{\|\pi(\mathbf{X}_j; \xi) - \mathbf{x}_j\|^2}{\sigma_{\text{px}}^2} + \frac{(\phi - \phi_{\text{imu}})^2 + (\theta - \theta_{\text{imu}})^2}{\sigma_{\text{att}}^2}, \quad (4)$$

solved by the Levenberg-Marquardt method [25], [26] from two starting points, the closed-form pose and its mirrored counterpart, so that the planar ambiguity [14], [15] is examined explicitly. The second term is a prior on roll and pitch from the inertial sensor ($\sigma_{\text{att}} = 2^\circ$). Fixing the attitude rigidly instead had failed for close markers, where an attitude error of one degree moves the predicted corners by several pixels.

A. A cube law for the weak direction

Consider one marker seen head-on at range r with focal length f and rotated by a small angle α about its vertical axis. Its two vertical edges differ in apparent height by $\Delta h \approx fs^2 \sin \alpha / r^2$, which is the only first-order observation of the rotation and has a standard deviation of $2\sigma_{\text{px}}$. A camera that misjudges the orientation of the marker places itself on the wrong point of a circle of radius r around it, so the position uncertainty across the line of sight is

$$\sigma_{\perp} \approx r \sigma_{\alpha} = \frac{2 \sigma_{\text{px}} r^3}{f s^2}. \quad (5)$$

Fig. 6 compares (5) with the covariance that the estimator reports. The attitude prior constrains rotation about the two horizontal axes and hence the vertical position, which improves from 2.75 to 0.32 m at 9 m; it says nothing about rotation about the vertical axis, so the horizontal uncertainty is unchanged. Only a heading reference or a second marker at another bearing can remove it. Fixes are therefore gated at 10 m and 45 pixels,

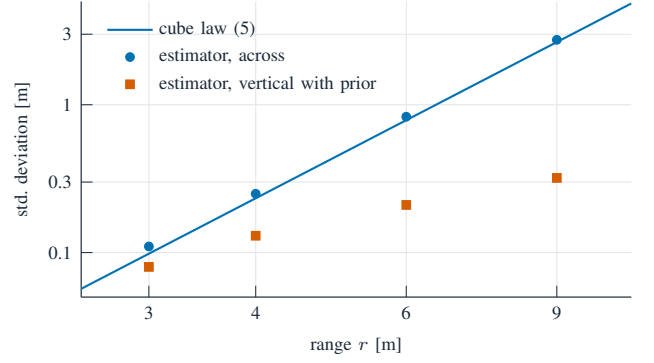


Fig. 6. Uncertainty of a single-marker fix against range ($\sigma_{\text{px}} = 0.5$ pixels). Across the line of sight it follows the cube law with or without the attitude prior; the prior bounds only the vertical component.



Fig. 7. Forward camera during validation. The near board is detected and used; the board in the background is below the size gate and is ignored.

and each is published with its covariance $(\mathbf{J}_r^T \mathbf{J}_r)^{-1}$, which a unit test confirms against the scatter of repeated noisy fits to within 30 %.

B. Validation

In Stonefish the vehicle moved from side to side 4 to 5 m in front of one board (Fig. 7) while the fixes were compared with ground truth (Table II). Facing the board, the mean error is 0.05 m; parallel to it, with the board seen obliquely, it doubles. The reported uncertainty is consistent in all runs, and the method tolerates the poor attitude reference of the default scenes (5.5° in roll and pitch), since close markers outweigh the prior. The errors are nearly constant and not scattered, which a processing delay of about 0.15 s would explain; this has not been separated by a stationary test, and nothing has been measured beyond 5 m.

VIII. CORAL PERCEPTION AND MAPPING

The pipeline has four stages: a detector that returns a family, a confidence and a mask for each coral; a tracker; range estimation from a single moving camera; and a map of landmarks with covariances. Two interchangeable detectors were built: a YOLO-family instance segmentation network [19] and an oracle that reads the segmentation camera of the simulator, against which the three geometric stages were developed.

TABLE II
MARKER LOCALIZATION AGAINST GROUND TRUTH, ONE BOARD AT ABOUT
4 m, SINGLE RUNS OF 40 s

Run	Mean	95 %	Heading	In 2 σ
Level camera, ideal IMU	0.055 m	0.070 m	0.42°	100 %
Level camera, noisy IMU	0.121 m	0.255 m	0.51°	95 %
Pitched, parallel to board	0.102 m	0.123 m	0.95°	100 %
Pitched, facing board	0.052 m	0.059 m	0.57°	99 %



(a) Downward camera (b) Forward camera
Fig. 8. Training labels generated automatically from the simulator.

A. Automatic labelling

The segmentation camera gives a mask for each object but identifies objects by numbers whose order the scene file does not state. The correspondence is discovered from the images: points sampled on the surface of each coral mesh are projected into the segmentation image with the true vehicle pose, and each casts a vote for the object number at its pixel. A number is assigned to a coral when at least 70 % of its votes come from that coral, or at least 40 % when coral and number are each other's first choice. The second rule was needed for table corals, whose share is diluted by smaller corals hidden under their canopy. Training images are additionally perturbed in the ways water alters an image: attenuation of red, a blue-green veil, blur and noise (Fig. 8).

B. Tracking and ranging

Detections are associated by box overlap in two passes in the manner of ByteTrack [17], with the Hungarian algorithm [27]. Range comes from three sources. The ray through the base of a detection meets a seabed model, which gives a position from the first frame. As the vehicle moves, rays with origins $\mathbf{o}_i$ and directions $\mathbf{d}_i$ are triangulated by weighted least squares,

$$\left(\sum_i w_i \mathbf{P}_i\right) \hat{\mathbf{c}} = \sum_i w_i \mathbf{P}_i \mathbf{o}_i, \quad \mathbf{P}_i = \mathbf{I} - \mathbf{d}_i \mathbf{d}_i^\top, \quad (6)$$

whose inverse information matrix is the covariance. For two rays at range r spanning an angle β , the uncertainty along the viewing direction is $r\sigma_\alpha/(\sqrt{2}\sin(\beta/2))$: at 3 m and $\sigma_\alpha = 1^\circ$ it is 0.85 m at $\beta = 5^\circ$ and 0.14 m at 30° , so a triangulated position is accepted only from at least three rays spanning 5° or more. Finally, a coral cut by the image edge, the usual case for the downward camera, is placed at the middle of the footprint that its mask outline accumulates on the seabed model. A point of the outline at height h seen at an angle γ from the vertical is displaced by $h \tan \gamma$, which is of the order

TABLE III
CORAL MAPS FROM SURVEY FLIGHTS, TRUE VEHICLE POSE, ONE FLIGHT
EACH

Area, camera	Detector	Mapped	Class	Error
Training, down	oracle	50/53	48/50	0.20 m
Training, down	learned	50/53	48/50	0.20 m
Training, forward	oracle	53/58	52/53	0.13 m
Training, forward	learned	52/58	50/52	0.18 m
Held-out, down	oracle	11/13	11/11	0.23 m
Held-out, down	learned	11/13	9/11	0.24 m
Held-out, forward	oracle	3/15	2/3	0.32 m
Held-out, forward	learned	6/15	6/6	0.14 m

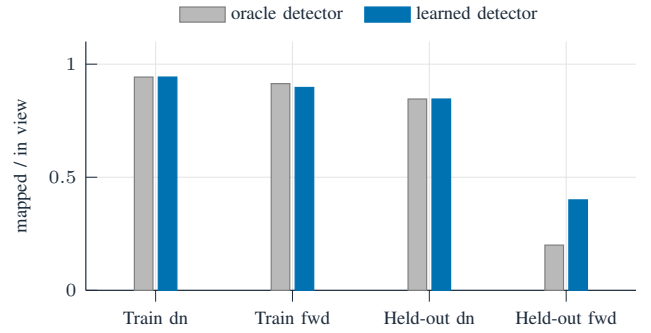


Fig. 9. Share of the corals in view that were mapped. The learned detector matches the oracle except with the forward camera on the held-out strip, where both fail.

of the errors measured below and the principal limit of the method.

Tracks are attached to landmarks by footprint overlap, by distance, or by a Mahalanobis gate, and triangulated estimates are fused by their information. A landmark is published when it has been observed five times and its largest standard deviation is below 0.5 m.

C. Results

Survey flights at 3 m above the seabed were recorded and replayed through the pipeline. A first detector, trained on one small survey, had learned individual corals and not families: on unseen corals it named three of seven correctly. For the second iteration the reef of 70 corals was divided into a training area of 53 corals and a held-out southern strip of 17, and the forward camera was pitched 20° down. Table III and Fig. 9 give the results. From the downward camera the learned detector reproduces the oracle's map exactly in count and error, and on the held-out strip it maps the same eleven of thirteen corals and names the family of nine. Sensitivity checks on the first survey, with 0.1 m of noise on the vehicle position, and with a flat seabed in place of the terrain model, changed the mean error by at most 0.05 m. The forward camera performs poorly on the held-out strip with either detector, so the fault does not lie in detection: tracks broke continually on that flight, and the cause has not been found.

TABLE IV
CLEAN SIMULATOR STARTS BEFORE AND AFTER THE CORRECTION, FOR
THREE WAYS OF FILLING NEWLY ALLOCATED MEMORY

Fresh memory contains	Before	After
Whatever the process left (normal)	3 of 20	10 of 10
A junk byte pattern	0 of 6	5 of 5
Zeros, recycled blocks all ones	2 of 6	4 of 4

IX. GUIDANCE

Autonomous flight required no change to the controller. A guidance node turns a route into velocity commands: the horizontal setpoint is proportional to the position error (0.6 s^{-1}), limited to 0.3 m/s , rotated into the body frame, and scaled down while the heading error is large, so that the vehicle turns almost on the spot at the end of a leg and the cameras do not sweep across the scene. A survey is a back-and-forth pattern whose waypoints follow the terrain at a fixed height above the shallowest ground within 1 m. At 3 m the downward camera covers a strip 5.0 m wide, so legs 2.5 m apart show every point from two sides, as the footprint method requires. Three surveys of 136 to 737 waypoints, the longest with 805 m of track, were flown from start to finish without intervention.

X. VERIFICATION AND A SIMULATOR FAULT

The arithmetic of every package is kept in modules that do not import ROS 2 and is covered by 156 unit tests that run in under a minute without the simulator. Nodes are then tested from their launch files with rendered images, which exposed two localization faults before any simulator run.

On some starts the simulator reported speeds above a thousand metres per second and the vehicle froze or left the scene. Four explanations were set aside: a start inside a coral (the hull was 0.68 m clear of the nearest mesh), an overloaded graphics card, a thread race, and a large commanded thrust (every command sent before the first faulty sample was zero). The first had appeared confirmed when moving the start position was followed by a series of clean starts; the failures returned a day later, in 17 of 20 starts. A fault that occurs at random will sooner or later appear to respond to any change.

A result that varies between identical runs of a deterministic program points to memory that is read before it is written. This was tested by filling freshly allocated memory deliberately (Table IV): with a junk pattern every start failed. The source of Stonefish shows the reason. The constructor of the simplified thruster model does not initialise the two setpoints from which force and torque are copied at every physics step, and the ROS 2 interface writes them only when a command has been received. For the first steps each thruster applied whatever its memory held. The interface now zeroes the setpoints when the scene is built. Guards were added so that an implausible state aborts a mission and is never recorded.

XI. LIMITATIONS

No experiment was carried out on the hardware of the third version; the simulated thrusters are ideal and the water is

still. The controller, guidance and perception used ground-truth state, and the marker localization has not yet closed a loop. All measurements are single runs, and the conclusions rest only on large differences. The detector was trained and evaluated in one simulated reef, with one family absent from its training data, and its figures do not predict performance in the sea. The roll loop has little margin, and the identified plant has not been compared with the vehicle.

XII. CONCLUSION

The project now has a sealing method that holds, a simulation the whole team can run, closed-loop control in six degrees of freedom and the first components of autonomy. Two lessons recur. Faults were found by measurement and not by inspection: grey corals were a property of light in water, gains from an assumed mass were several times too weak, and a vehicle thrown out of the scene had touched nothing. And an estimate is only useful with an honest uncertainty: the marker fixes and coral positions carry covariances that were checked against the actual scatter. The step that remains is to close the loop on estimated instead of true state, by fusing the marker fixes with an innovation gate and adding a velocity source, first in simulation and then in the pool.

ACKNOWLEDGMENT

ROVIO is the work of the RMI club. I thank Kuhan Anandh, who made the first Stonefish set-up in the team, Aditya Ganta, who helped with the frame conversion of the model, Yagishvar, who advised on teleoperation, and the mechanical and electronics groups who designed and built the vehicle. The simulation rests on Stonefish and its ROS 2 interface by Patryk Cieślak.

REFERENCES

- [1] M. M. M. Manhães, S. A. Scherer, M. Voss, L. R. Douat, and T. Rauschenbach, "UUV simulator: A Gazebo-based package for underwater intervention and multi-robot simulation," in *OCEANS 2016 MTS/IEEE Monterey*, 2016, pp. 1–8.
- [2] M. M. Zhang, W.-S. Choi, J. Herman, D. Davis, C. Vogt, M. McCarrin, Y. Vijay, D. Dutia, W. Lew, S. Peters, and B. Bingham, "DAVE aquatic virtual environment: Toward a general underwater robotics simulator," in *2022 IEEE/OES Autonomous Underwater Vehicles Symposium (AUV)*, 2022, pp. 1–8.
- [3] E. Potokar, S. Ashford, M. Kaess, and J. G. Mangelson, "HoloOcean: An underwater robotics simulator," in *2022 International Conference on Robotics and Automation (ICRA)*, 2022, pp. 3040–3046.
- [4] P. Cieślak, "Stonefish: An advanced open-source simulation tool designed for marine robotics, with a ROS interface," in *OCEANS 2019 - Marseille*, 2019, pp. 1–6.
- [5] D. Akkaynak and T. Treibitz, "A revised underwater image formation model," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 6723–6732.
- [6] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. Chichester: John Wiley & Sons, 2011.
- [7] G. Antonelli, *Underwater Robots*, 4th ed. Cham: Springer, 2018.
- [8] L. Meier, D. Honegger, and M. Pollefeys, "PX4: A node-based multi-threaded open source robotics framework for deeply embedded platforms," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 6235–6240.
- [9] K. J. Åström and T. Hägglund, *Advanced PID Control*. Research Triangle Park, NC: ISA - The Instrumentation, Systems, and Automation Society, 2006.
- [10] T. A. Johansen and T. I. Fossen, "Control allocation: A survey," *Automatica*, vol. 49, no. 5, pp. 1087–1103, 2013.

- [11] E. Olson, "AprilTag: A robust and flexible visual fiducial system," in *2011 IEEE International Conference on Robotics and Automation*, 2011, pp. 3400–3407.
- [12] S. Garrido-Jurado, R. Muñoz-Salinas, F. J. Madrid-Cuevas, and M. J. Marín-Jiménez, "Automatic generation and detection of highly reliable fiducial markers under occlusion," *Pattern Recognition*, vol. 47, no. 6, pp. 2280–2292, 2014.
- [13] D. B. dos Santos Cesar, C. Gaudig, M. Fritsche, M. A. dos Reis, and F. Kirchner, "An evaluation of artificial fiducial markers in underwater environments," in *OCEANS 2015 - Genova*, 2015, pp. 1–6.
- [14] G. Schweighofer and A. Pinz, "Robust pose estimation from a planar target," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, no. 12, pp. 2024–2030, 2006.
- [15] T. Collins and A. Bartoli, "Infinitesimal plane-based pose estimation," *International Journal of Computer Vision*, vol. 109, no. 3, pp. 252–286, 2014.
- [16] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *2016 IEEE International Conference on Image Processing (ICIP)*, 2016, pp. 3464–3468.
- [17] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in *Computer Vision - ECCV 2022*, ser. Lecture Notes in Computer Science. Springer, 2022, pp. 1–21.
- [18] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [19] G. Jocher, J. Qiu, and A. Chaurasia, "Ultralytics YOLO," <https://github.com/ultralytics/ultralytics>, 2023.
- [20] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge: Cambridge University Press, 2004.
- [21] J. Civera, A. J. Davison, and J. M. M. Montiel, "Inverse depth parametrization for monocular SLAM," *IEEE Transactions on Robotics*, vol. 24, no. 5, pp. 932–945, 2008.
- [22] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30.
- [23] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot operating system 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, 2022.
- [24] T. Moore and D. Stouch, "A generalized extended Kalman filter implementation for the Robot Operating System," in *Intelligent Autonomous Systems 13*, ser. Advances in Intelligent Systems and Computing. Cham: Springer, 2016, pp. 335–348.
- [25] K. Levenberg, "A method for the solution of certain non-linear problems in least squares," *Quarterly of Applied Mathematics*, vol. 2, no. 2, pp. 164–168, 1944.
- [26] D. W. Marquardt, "An algorithm for least-squares estimation of non-linear parameters," *Journal of the Society for Industrial and Applied Mathematics*, vol. 11, no. 2, pp. 431–441, 1963.
- [27] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.